

COLE A. MURRAY

coeam2u@gmail.com | 847-460-2068 | www.linkedin.com/in/cole-murray-184679367 | colemurray.dev

EDUCATION

University of Illinois at Urbana-Champaign, Grainger College of Engineering

GPA: 3.75/4.00

Bachelor of Science in Computer Engineering, Hoeft Technology and Management Minor

Expected Graduation May 2029

Related Coursework:

Digital Systems Laboratory, Analog Signal Processing, Computer Systems & Programming, Introduction to Computing, Introduction to Electronics, Linear Algebra with Computational Applications, Discrete Structures, Introduction to Differential Equations, Physics: Electricity and Magnetism, Physics: Mechanics

Awards: Grainger Engineering Visionary Scholarship, Schaumburg Business Association Scholarship Winner, Illinois State Seal of Biliteracy (Spanish)

Certifications: Anthropic: Claude Developer Platform 101, Intro to Model Context Protocol, Claude Code in Action

Involvements: Hoeft T&M Program, Phi Gamma Nu Professional Business Fraternity, Illinois Space Society, Illini Run Club

TECHNICAL SKILLS

Programming Languages: C/C++, Python, TypeScript, SystemVerilog, Java, SQL, ReactNative, JavaScript, Assembly (LC-3)

Frameworks/Tools: Microsoft Azure, Microsoft Entra, Git, Snowflake, REST API, Supabase, MCP, Supabase, Cursor, Vivado

WORK EXPERIENCE

Nerdio

Skokie, IL

Software Engineer Intern

May 2026 – August 2026

- Architected a **TypeScript MCP server** integrating Microsoft Graph and Azure ARM APIs to audit privileged Entra ID / Azure RBAC access across a **60+ tenant multi-cloud identity platform**; 12 tools, OAuth app-only auth, and Azure Container Apps hosting with Entra-authenticated HTTP transport
- Built an authenticated **LLM/API proxy** so internal apps can call Claude without exposing the Anthropic API key; validates Entra ID JWTs, injects a server-side key, proxies the Anthropic API, and enforces per-user allowlists, model allowlists, and rate limits; deployed to Azure App Service with Key Vault secrets and structured audit logging
- Engineered a TypeScript **Azure Function App** multi-destination **webhook delivery platform** that replaced scattered automations with a queue-backed routing system to Teams, Azure Communication Services Email, or HTTPS webhooks, with an append-only Table Storage audit log; Entra Easy Auth and a custom RBAC admin dashboard on Static Web Apps, with GitHub Actions OIDC CI/CD

Solutions Engineer Intern

May 2025 – August 2025

- Designed and built internal tools to support product, customer success, and operations teams
- Developed interactive dashboards and apps to visualize product usage, customer interaction, and web traffic
- Created large scale data transformations to deliver actionable datasets to multiple departments
- Built and implemented multiple automation workflows, including a real time automated reply system for the company help forum and an email notification system for incoming support tickets
- Gained familiarity leveraging AI tools such as Cursor and Copilot 365 to increase and optimize my output

PERSONAL PROJECTS

First10 | Garmin Connect IQ, Monkey C, Embedded Programming, Constrained Hardware

- Built a Garmin **wearable app** that keeps an alarm active until a multi-sensor Awake Score (steps, sustained motion, and heart rate rise above a resting baseline) is passed, with sensors and alerts gated to the alarm phase
- Designed around Connect IQ limits: foreground-only vibration/tone and high-power CPU while open, keeping the AMOLED screen off and sensors idle during sleep, then enabling **HR and accelerometer sampling** when alarm fires
- Implemented difficulty presets, an HR-gated Hard mode resistant to wrist shakes, snooze limits, escalating alerts, and safety dismiss; added capability-checked wrappers, HR-unavailable fallback, and unit tests for scoring and hold timing

Simple Microprocessor | SystemVerilog, Xilinx Vivado, Spartan-7 FPGA

- Designed a 16 bit RISC processor (11 instruction ISA) in SystemVerilog, including a register file, ALU, and multiplexer based shared datapath bus. Tested in **behavioral simulation** and on hardware, debugged with Vivado's Integrated Logic Analyzer
- Built an **FSM control unit** for the fetch-decode-execute cycle, adding multi-cycle wait states for block RAM with no ready signal, and integrated on-chip BRAM and memory-mapped I/O so programs could read switches and drive hex displays with ordinary load/store instructions